



Career in Database Administration

The explosion of information and the evolution of Big Data has given rise to one of the most lucrative careers in the IT sector. Find out how you can ride this information bandwagon.

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‘Big Data’ is such a common buzzword these days, so let us tell you what the fuss is all about. In simple words, Big Data is data that is so diverse, fast changing and substantial that those who work with it must go beyond conventional technologies, skills and infrastructure to use it efficiently.

The boom in the IT industry has come in tandem with an upsurge in the amount of data associated with every organisation. Big Data is this ever-increasing volume and velocity of data that we have to deal with today. Over the past couple of decades, there’s also been a shift in the type of data stored, which now significantly comprises of social media, mobile data and machine data, apart from the run-of-the-mill text and numeric relational data.

According to Kamal Brar, Vice President, APAC at MongoDB (provider of commercial Open Source NoSQL database software), there’s high demand



Big Data getting bigger

for Big Data Management technology in India. The sheer volume of online data has increased in the past decade. In the early 2000s, the size of a traditional relational database was 7 TB. Today, individual users are clocking around 4-5 Petabytes of data.

The world’s information is doubling every two years. In its 2011 report, IDC estimated the size of the digital universe to be 1.8 Zettabytes. According to the study, by 2020 the world would generate 50 times the amount of information and 75 times the number of ‘information containers’ while information managing IT staff will grow by 1.5 times (approx).

The US Bureau of Labor Statistics reports that by 2020, computer jobs around the world are expected to increase at a rate of 14.3 per cent – at a faster rate than any other occupation. Computer occupa-

tions cover a myriad of profiles such as information research scientists, computer system analysts, computer programmers, network system administrators, web developers and database administrators, to name a few.

Background

Today, an increasing number of organisations are gathering data about their consumers and competitors to gain useful market insights that can be translated into a business edge. Big Data is only going to get bigger, and soon it will outstrip the capacity of the people being trained to manage it. And the job doesn’t simply involve crunching numbers, but also analysing (transforming and processing) data in a way that renders it useful. While the industry has created jobs, it needs people who can find innovative ways to efficiently utilise this humongous amount of data.

How data has changed in the past decade

“What has changed is how we manage and mine that data. Traditionally a lot of data that we had was transactional in nature. Today, we store data which is very different in the way it’s captured – like social media and machine-to-machine data”, Kamal elucidates. This sensory-based data is being captured not on an hourly basis, but every millisecond.

The data today has multiple dimensions, so organ-

The scale of data

- 1 Bit = Binary Digit
- 8 Bits = 1 Byte
- 1024 Bytes = 1 Kilobyte
- 1024 Kilobytes = 1 Megabyte
- 1024 Megabytes = 1 Gigabyte
- 1024 Gigabytes = 1 Terabyte
- 1024 Terabytes = 1 Petabyte
- 1024 Petabytes = 1 Exabyte
- 1024 Exabytes = 1 Zettabyte
- 1024 Zettabytes = 1 Yottabyte
- 1024 Yottabytes = 1 Brontobyte
- 1024 Brontobytes = 1 Geopbyte



Investment in Digital Universe (IDC Digital Universe Study 2011)